SAMIUL BARI

Ph.D. Student | Electrical and Computer Engineering

Address: 3500 Sun Bowl Dr, Apt #86, El Paso TX 79902 | Phone: (915) 203-2715 | sbari@miners.utep.edu

Personal Website | Laboratory

RESEARCH PROFILE

Ph.D. researcher specializing in efficient artificial intelligence, multimodal deep learning, and AI hardware acceleration. Research focuses on developing computationally efficient and hardware-aware AI models through model compression and optimization for resource-constrained environments. Expertise includes biomedical signal processing, time-series modeling, AI-augmented medical devices, and physiological digital twins, with an emphasis on real-time inference, intelligent health monitoring, and personalized healthcare applications.

EDUCATION

Ph.D., Electrical and Computer Engineering | University of Texas at El Paso | 2025–Present

Advisor: Md Maruf Hossain Shuvo | Edge and Biomedical Intelligence (EBMI) Laboratory

B.S., Biomedical Engineering | Military Institute of Science and Technology | 2020–2024

Bangladesh University of Professionals | GPA: 3.54/4.00

RESEARCH AND TEACHING EXPERIENCE

Ph.D. Research Associate, EBMI Laboratory, UTEP | *2025–Present*

Investigate efficient AI methods for edge and biomedical applications including deep learning model compression, efficient hardware acceleration and AI embedded healthcare systems. Develop and evaluate physiological learning systems and explore hardware-aware inference on FPGA platforms.

Teaching Assistant, Department of Electrical and Computer Engineering, UTEP | *Fall 2025–Present*

Courses: Capstone Senior Projects and Electrical Circuits II

Support circuit analysis and laboratory instruction, student troubleshooting, assignment evaluation, and senior-design project development and technical presentations.

ONGOING RESEARCH PROJECT

QuBeT — ECG-Based Glucose State Sensing | *Ongoing*

Developing a quantile beat transformer to classify glucose states from single-lead ECG. The proposed design includes mask modeling of heart beats, sinusoidal time encoding and CNN beat tokenizer. Physiological assumptions and predictive performance are under investigation.

SELECTED RESEARCH PROJECTS

ExerTwin-T1D — Exercise-Aware Digital Twin for Type 1 Diabetes | *Under review, IEEE JBHI*

Developed a physiologically grounded digital twin that incorporates exercise-related insulin-sensitivity dynamics into glucose–insulin kinetics model. Evaluated individualized model calibration and counterfactual exercise, meal, and insulin scenarios.

Swin-VSA — Gastrointestinal Image Classification | *Under review, IEEE JBHI*

Developed a Varied-sized Window Attention-based Swin Transformer system for gastrointestinal image classification and evaluated various AI model compression techniques for efficient deployment by reducing the model size by approximately 74% and computational load by 59% relative to the baseline.

Severity Classification of Diabetic Retinopathy from Smartphone Fundus Images

Developed and evaluated a modified DenseNet-121-based five-class grading framework with retinal-image preprocessing and transfer learning. Reported 95.2% macro-F1 and assessed smartphone-acquired images with per-prediction confidence scores.

FPGA Spiking Neural Network Accelerator — Xilinx PYNQ-Z1

Implemented a quantized leaky integrate-and-fire neural network accelerator on FPGA and evaluated MNIST inference, reporting 89.56% accuracy and 0.236 ms latency per image.

CAPSTONE PROJECT MENTORSHIP

Edge AI System for Real-Time Obstructive Sleep Apnea Detection

Developed a wearable, single-lead ECG-based system that integrates signal acquisition, transformer-based apnea detection, and Raspberry Pi 4 edge inference. Achieved 96.7% binary detection accuracy and 80.0% four-class severity classification accuracy using the OSASUD dataset.

TECHNICAL SKILLS

AI/ML: PyTorch, TensorFlow, transformers, quantization, pruning, computer vision, time-series analysis

Hardware & Programming: FPGA, Vivado/Vitis, Verilog, Python, C/C++, MATLAB, Cadence Virtuoso, OpenVINO

Biomedical: ECG, EEG, EMG, PPG, physiological modeling, signal processing, wearable sensing